Leo Selker

selker.leo@gmail.com
503 560 9593

Education

- **University of Washington** — Seattle, WA
  MS, Applied and Computational Mathematics. GPA 3.89/4 — *June 2022*

- **Pomona College** — Claremont, CA
  Bachelor of Arts in Mathematics, cum laude. Computer Science minor. GPA 11.53/12[1] — *May 2017*
  - Awarded distinction on senior thesis (mathematics): *Möbius Inversion: From Posets to Categories*, advised by Dr. Vin de Silva.
  - Benjamin J. Levy Memorial Prize in Mathematics (faculty chosen).

Research Experience

- **Center for Effective Global Action, Global Policy Lab, UC Berkeley** — Berkeley, CA
  Data Scientist, supervised by Dr. Joshua Blumenstock. — *2022-Present*

  Operated in Joshua Blumenstock's research group in both a research and operations capacity. Focus has been using statistics and machine learning to study the allocation of funds for social protection programs.

  Research includes pricing optimal allocation of social-protection funds (see below) and investigating methods for estimating household-level standard of living in Malawi and Haiti.

  Operationally supported several cash aid programs (Malawil Haiti, Bangladesh). Developed software, built machine learning models, and managed relationships with collaborators, including international governments and NGOs.

  Supervised several research assistants and a summer intern.

- **University of California, Berkeley** — Berkeley, CA
  Research Associate, supervised by Dr. David Romps — *2020*

  Developed algorithms for mapping daily weather data between climates. Used statistical and numerical approaches to interpret non-stationary observations. Operated semi-autonomously in an academic research group. Python, Xarray, Numpy/Scipy.

- **Claremont McKenna College** — Claremont, CA
  Research Assistant, supervised by Dr. Sam Nelson — *2015*

  Knot theory/topology research. Developed and evaluated algebraic knot invariants. Coauthored paper (see Publications) and presented our work at the Knots in Washington conference (Dec 2015).

Publications, Independent Study, and Presentations

- Sahoo, Blumenstock, Niehaus, **Selker**, and Wager. 2025. What Would It Cost to End Extreme Poverty? NBER Working Paper #34583.

- Senior Thesis in Mathematics (2016-2017) *Möbius Inversion: From Posets to Categories*, advised by Dr. Vin de Silva. Text available on github or upon request.

- Kaestner, A., Sam Nelson, & **Leo Selker**. (2016). *Parity Biquandle Invariants of Virtual Knots*. Topology and its Applications, 209, 207–219. http://doi.org/10.1016/J.TOPOL.2016.06.010. Note: authors are ordered alphabetically.

- **Presented at *Knots in Washington* conference.** Title: *Parity Biquandle Invariants of Virtual Knots*, on work published in similarly-titled paper (above). George Washington University, December 2015.

[1] The college recommends dividing by 3 to convert to a four-point scale. My GPA equivalent is therefore 3.84/4.

Other Experience

- **NewGrid, inc.** Boston, MA (remote)
  Software Engineering Consultant *2020 - 2021*

  Helped develop power grid topology optimization software. Helped with R&D and implementation of new heuristics to predict beneficial reconfigurations. Applications include improving grid throughput, maintaining robustness under contingencies, and increasing the viability of renewable installations. Python.

- **Google** San Francisco, CA
  Software Engineer; Authentication and Authorization Infrastructure *2017 - 2019*

  Software engineer level III → IV

  Designed, built, and maintained authentication and authorization infrastructure. Worked with a variety of protocols (OAuth, OIDC, RISC, FIDO, etc) and associated security considerations. Used SQL and built pipelines to analyze server traffic data. Java, C++, SQL, and other languages.

- **Ayasdi** Menlo Park, CA
  Software Engineering Intern *2016*

  Used classical statistical tools as well as techniques from information theory to implement supervised and unsupervised feature selection algorithms, and integrated them with Ayasdi's machine intelligence and topological data analysis platform. Java, Python.

- **Pomona College Department of Mathemnatics** Claremont, CA
  Teaching Assistant *2015-2017*

  Worked as a TA for combinatorics, abstract algebra, and algebraic topology special-topics seminar.

Selected Coursework

Data analysis/machine learning; Two semesters of real analysis; applied complex analysis; analysis of stochastic processes; network optimization; numerical analysis of differential systems; dynamical systems; differential equations; operations research; statistics; applied algebraic topology.

Professional and academic references are available upon request.